Anti-Corruption Base Intelligence

Design Overview-V-2

Plantel: Centro Universitario de los Valles

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1. Introduction

This document provides an overview of the design of the "Anti-Corruption Base Intelligence (AcBi)" mobile application. The system is designed as a multi-platform (Android and iOS) system for government entities to proactively detect corruption patterns in public administration.

The system architecture prioritizes resilience and security through a local-first approach and offline capabilities. The core functionality is delivered through five integrated critical modules: Data Listener, Network Analysis, Process Tracking, Relational Time Lines (RTL), and the Time Paradox dashboard.

From a technical perspective, AcBi implements security protocols. These include end-to-end encryption (E2EE), a digital chain of custody based on SHA-512 hashes, and certified timestamps, all aimed at preserving the integrity of investigations.

2. System Objectives

The technical purpose of AcBi is to establish a forensic intelligence infrastructure for identifying administrative anomalies. The design focuses on achieving the following operational milestones:

- Award Monitoring: Automate the detection of irregularities in government tenders and contracts by analyzing input data.
- Relationship Analysis: Map beneficiary networks to uncover illicit connections and hidden power structures.
- Temporal Inconsistencies Detection: Identify "time paradoxes" where administrative milestones do not comply with the required logical or legal sequence.

Design Overview

The diagram illustrates how AcBi's five core modules interact in a pipeline flow. M1: Data Listener acts as the entry point, ingesting raw data and feeding it simultaneously into M2: Network Analysis and M3: Process Tracking, which analyze relationships and monitor procedural activity independently. Both modules converge into M4: Relational Timelines, which cross-references findings to construct a chronological map of events and connections. All outputs then consolidate into M5: Time Paradox Dashboard, which synthesizes the full investigative picture and delivers it as visualizations and alerts to the end user — the detective. Table 1.1 Design Overview.

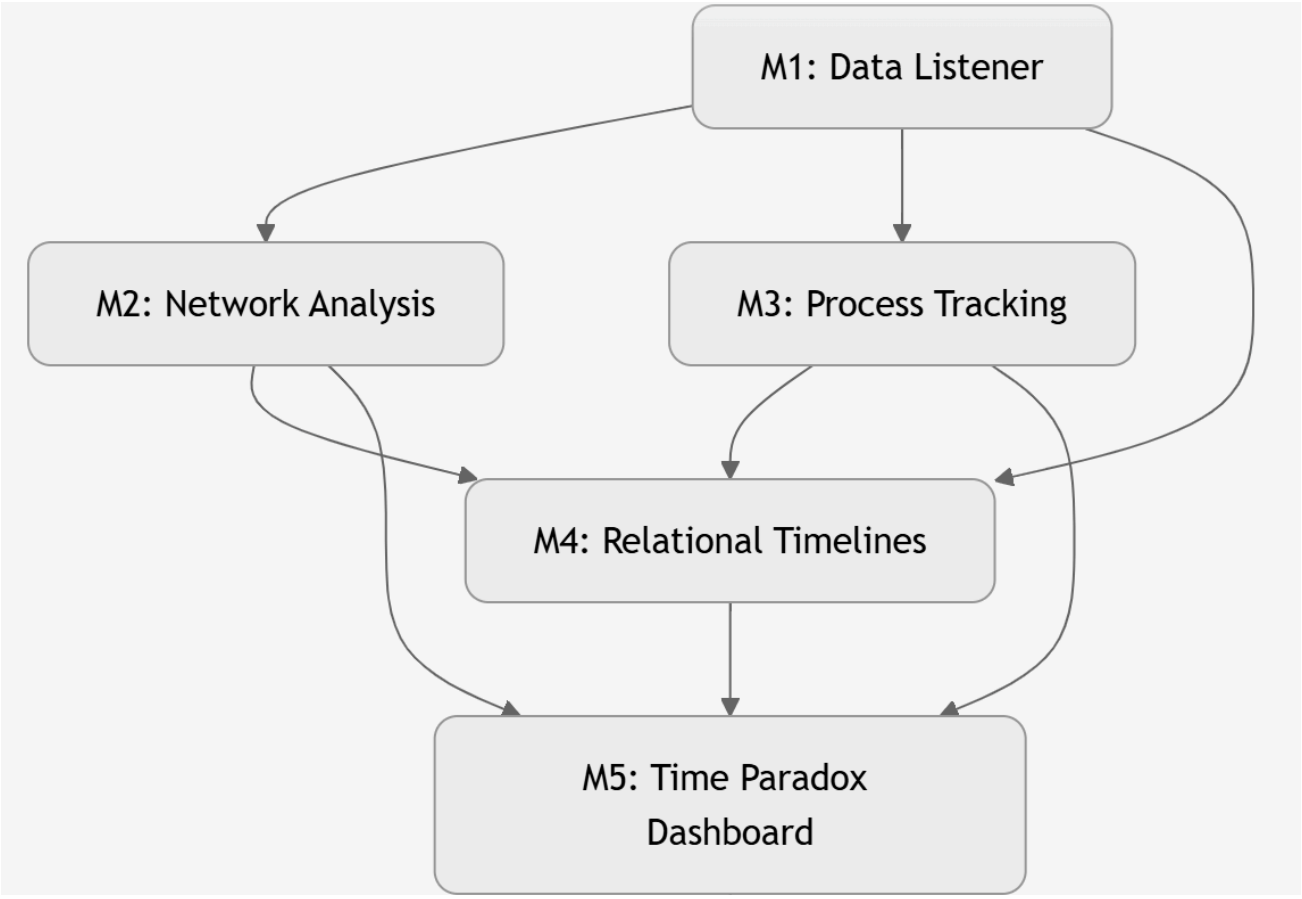

Table 1.1 Design Overview